Hands on Activity No. 4.3	
Sorting and Searching Array	
Course Code: CPE007	**Program:** Computer Engineering
Course Title: Programming logic and design	**Date Performed: 9/16/2025**
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6. Output

- Screenshot of Code(Readable):
- Output of Code(label and compile ALL possible outputs):
- Short Analysis(min 5 sentences)

7. Supplementary Activity

1.

MONDAY.cpp

```
#include <iostream>
using namespace std;

int main() {
    string days[] = {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};
    int num;

    while (true) {
       cout << "Enter a number (0-6): ";
    cin >> num;

        if (num >= 0 && num < 7) {
            cout << days[num] << endl;
        } else {
            cout << "Error, no such day." << endl;
            break;
        }
    }

    return 0;
}
```

Explanation: First, create a string array to Identify the value of the days of the week (Sunday to Saturday). Next, put an integer variable to serve as the identifier. Then, use a **WHILE (TRUE)** So the program continues running until the correct input is given. Inside, use **cout** to prompt the user to “enter a number” and **cin** to read the integer input. The **IF** Statement will check if the number is between 0 and 7. If the input is valid, it will display the corresponding day from the string array. If the input exceeds 7, the program will get an error, while the **else** will print the “Error, no such day” then use break to stop.

OUTPUT:

```
C:\Users\hanz\OneDrive\Docu
Enter a number (0-6): 1
Monday
Enter a number (0-6): 2
Tuesday
Enter a number (0-6): 3
Wednesday
Enter a number (0-6): 4
Thursday
Enter a number (0-6): 5
Friday
Enter a number (0-6): 6
Saturday
Enter a number (0-6): 9
Error, no such day.

--------------------------------
Process exited after 8.397 seconds with return value 0
Press any key to continue . . .
```

2:

```
#include <iostream>
#include <string>
 using namespace std;

 int main () {
     const int SIZE = 8;
     string board [SIZE][SIZE];

     for (int i = 2; i < 6; i++) {
         for (int j = 0; j < SIZE; j++) {
         board [i][j] = " ";
             }
      }

     string backRank [8] = {"R", "N", "B", "Q", "K", "B", "N", "R"};

     for (int j = 0; j <SIZE; j++) {
         board[0][j] = backRank[j];
         board[1][j] = "p";
         board[6][j] = "p";
         board[7][j] = backRank[j];
      }

      for (int i =0; i <SIZE; i++) {
         for (int j = 0; j <SIZE; j++) {
             cout<< board [i][j] <<" ";
          }
         cout <<endl;

      }

     return 0;

 }
```

Explanation: First, put a const int SIZE, the size will be the value 8. Next, the stringboard makes it 2D array. Then create a for loop this will count from 2 to 5 and will put a space, Then the string backRank will store the order of chess. Then another for to repeat 8 times [SIZE]
And then use another loop to print every square then use cout endl to stop the code.

```
C:\Users\hanz\OneDrive\Docu
R N B Q K B N R
p p p p p p p p




p p p p p p p p
R N B Q K B N R

--------------------------------
Process exited after 2.149 seconds with return value 0
Press any key to continue . . .
```

- Screenshot of Code:
- Output of Code (label and compile ALL possible outputs):
- Short Analysis(min 5 sentences):

8. Conclusion

I learned how to use arrays, loops, and conditions to build programs, like using while(true) for input checking and 2D arrays to create a chessboard layout The steps were clear and helped me understand how to build logic gradually declaring variables, looping through arrays, and displaying outputs. This improved my skills in using 2D arrays and nested loops, and helped me practice formatting and organizing outputs properly. I think I did well, but I need to improve

my logic planning and debugging to be faster and more accurate. With more practice, And soon I hope I become more versatile at coding like I can make a code with easier and more presentable.
Guide in creating a conclusion: ● Summary of lessons learned ● Analysis of the procedure ● Analysis of the supplementary activity ● Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement? note: answer this in a paragraph form not bullet or per question.